

FT8 on 11 m quick-start (sdroxide — CB/11 m)

A short, task-oriented walkthrough for working **FT8 on the 11 m (27 MHz) citizens' band** with **sdroxide**: which frequency, how to set it up, how to read the decodes and how to answer a station. On 11 m the whole exchange follows the community's [WSJT-CB](#) conventions rather than the amateur FT8 ones — because that is who is on the band. For the reference detail behind each control see [USER_MANUAL.md](#) §3.2.8 and §3.2; for the why of this fork see the [README](#).

This is the CB/SWL fork of sdroxide. FT8 itself is upstream's; what this fork adds on 11 m is the band, the per-country channel plans, the WSJT-CB exchange and the country flags.

Nederlands: [ft8-11m-quickstart.nl.md](#). Français: [ft8-11m-quickstart.fr.md](#). Italiano: [ft8-11m-quickstart.it.md](#). PDF: [ft8-11m-quickstart.en.pdf](#).

Bold names such as **SETTINGS** and **Callsign** are buttons and fields exactly as they appear on screen. Settings > Radio is a menu path.

The frequency (read this first)

- **FT8 on 11 m is 27.265 MHz — channel 26** of the 40-channel grid (the same grid under CEPT, the FCC and the ACMA).
 - The band's **digital calling channel is channel 25 = 27.245 MHz**, which is where pressing **11M** lands. FT8 has its own dial just above it.
 - The mode is **USB**. CB voice is AM/FM, but the digital modes live on USB.
 - Everything happens inside one **15-second slot** — see *Watch the clock*.
 - The UK **27/81** plan has its own numbering and no channel 25; use the FT8 dial the band offers there.
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What you need

- **An SDR or radio sdroxide supports**: RTL-SDR, RX-888, Airtyspy HF+, SDRplay RSP, HackRF, ELAD, PlutoSDR, a **CAT rig** over serial + sound card, TCI, OpenHPSDR or SoapySDR.
- **A 27 MHz antenna** suited to your receiver.
- **sdroxide installed** (below).
- **An accurate clock**. FT8 will not decode if your computer's clock is off by more than about a second. Turn on automatic time sync (NTP).

Installing

- **Windows** — the installer (.msi) or the portable .zip: see the [Releases page](#).
- **Linux** — the **AppImage** (one file, chmod +x and run), the .deb, or the portable tarball.
- **macOS** — the .dmg.

You can also run sdroxide as a **server** and open it in a browser: `sdroxide --server`, then `http://localhost:4950`.

Part A — One-time setup

Everything here is saved under `~/.config/sdroxide-brown/`, so you do it once.

1. Pick your radio

Open **SETTINGS** and go to the **Radio** tab. Pick your interface (for example **RTL-SDR**, **RX-888** or **Airspy HF+**), then the **sample rate** and the **gain**. Changes apply right after **Apply / reconnect**.

- **Linux — USB receivers:** install the bundled udev rules, or sdroxide will see the device but fail to open it: `sudo cp 60-sdroxide-*.rules /usr/lib/udev/rules.d/ && sudo udevadm control --reload`, then replug.
- **Windows — RX-888:** bind the device to **WinUSB** once with [Zadig](#) — for **both** USB ids (04B4:00F3 and 04B4:00F1).

2. Fill in your details

Go to the **General** tab:

- **Callsign** — your CB identifier, for example 26AT715. The leading digits are the **CB country number** (026 = England), and that is what draws the flag beside a decode.
- **Locator / grid** — your Maidenhead grid. On 11 m it is used for the map and the great-circle paths, but it is **not transmitted**: CB stations carry no locator.
- **CB plan** — **World / freeband**, **CEPT/EU**, **Germany 80 channels**, **UK 27/81**, **USA** or **Australia**. It fixes the channels and the channel numbers.

3. Only to transmit: allow 11 m

11 m transmit is off by default — see *A note on transmitting*. To work stations, switch on **Allow transmit on 11 m (CB)** on the General tab and confirm the warning once. If you only want to listen, skip this: see Part B.5.

Part B — Listening first

1. Press **11M** on the band bar. The band opens on the digital calling channel.
 2. Open the **Band / Mode** popup and choose **FT8** from the **DIGITAL** row. The panadapter locks onto the digital sub-band and the FT8 panel appears under the waterfall.
 3. The **slot bar** fills once per 15-second turn. When it reaches the end the decoder speaks and the next turn begins — an empty list under a bar that is still filling is a turn that has not finished.
 4. Watch the **DECODES** list. Each row carries the SNR, the audio tone, the callsign, the **country flag**, the continent, the distance and the full message; CQ calls are highlighted. Use **Sort** (SNR / Dist / Country), **CQ only** and **New only** to thin it out. A badge (**DXCC / BAND / GRID / NEW / DUPE**) says what the row would be worth against your log.
 5. **Listening only?** Turn on **SWL mode**: every transmit control disappears and **REPLY** is greyed. You still get the full decode list and flags.
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Part C — Answering a station

1. Click a decode row to put your transmit audio on that station, or press its **REPLY** button. The sequencer fills in the exchange and starts transmitting on the opposite slot.
 2. To call yourself, press **CALL CQ**. **STOP QSO** ends the contact.
 3. The exchange is WSJT-CB's, one call at a time as free text — 26AT715 -07, 26AT715 R+05, 26AT715 RR73, 26AT715 73 — and with **no grid** in any of it.
 4. The **station card** shows the current step and a transcript of the exchange (your lines in gold, theirs in green).
 5. Nothing leaves the radio until you have allowed transmit (Part A.3).
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Watch the clock

FT8's turn is 15 seconds and both ends must agree where it begins. The station card shows **DT** — how far your clock sits from the stations you are hearing. It is grey while you are inside half a second, amber past that and pink past 1.5 s. Positive means you transmit early. A clock far enough out that nobody can decode you looks exactly like a dead band from your side, so it is the first thing to check when nobody answers. Turn on your system's automatic time sync.

Part D — Keeping a record

- **Export the decode list** to **CSV** or a *received-report* **ADIF** (the **SAVE** chip) — handy for a log of what was on the band.
- **Profiles (Settings > Profiles)** save a whole working setup under a name: dial and mode, gain, drive, the digital identity and the band stacks.
- **Audible alerts** for calls and for new countries or grids.

- In a listening log (**LOG** in SWL mode) you can keep a **SINPO/SIO** report for each reception rather than a QSO.
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Sending

- **CB is meant for sending *and* receiving, and in most countries it needs no licence** — the licence-free CEPT channels, at limited power. This fork treats 11 m as a full band.
 - **To send FT8 (or SSTV) you need a transceiver**, keyed by **VOX** (a sound-card rig — the program plays the audio to its mic input and the radio keys itself) or over **CAT**. A receive-only dongle — most RTL-SDR, RX-888 and Airspy HF+ setups — decodes the band beautifully but cannot transmit; for sending you need a transceiver behind it.
 - **To transmit, switch on Allow transmit on 11 m (CB)** on the General tab and confirm the note once. It is a switch only because the program's amateur-band lockout is generic and refuses every non-amateur allocation; CB is a separate radio service, so it is opted into rather than assumed.
 - The switch opens **11 m and nothing else** — the broadcast bands stay receive-only.
 - Keep to your country's channels, power and modes — ordinary CB courtesy.
Check the current rules where you are.
 - This fork is for the whole of CB — voice and the digital modes — alongside SWL and decoding. CB and amateur radio are neighbours on the same spectrum.
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Help and links

- [README](#) — the full overview of the fork.
- [USER_MANUAL.md](#) — the manual, control by control.
- [WSJT-CB](#) — the digital 11 m project this builds on.
- [Releases](#) — the latest build for every platform.

See you on 27.265! 73